

A Sunrise PEPPERED WITH DRONES



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When I started work on this story at the close of 2015, the USA's Federal Aviation Administration (FAA) was voicing serious concerns about the number of drones expected to be bought by people during the 2015 Christmas sales. Ranging from simple US\$ 20 toys to high-end quadcopters, FAA was expecting a million unmanned aerial vehicles (UAVs, aka drones) to be sold. The concern is palpable, because these new aircraft, if not flown responsibly, could cause a lot of trouble for airlines. So much so that FAA has launched a beta application called B4UFly, which helps drone users to abide by flying regulations, especially to stay away from prohibited zones where their little aircraft could cause harm to real big ones.

We are not sure how many drones really got sold over Christmas holidays, but it is evident that drones represent a real, solid trend. At one time, drones fell into two categories: either these were used in classified, military operations or were toys. Now their applications are more real-world.

In July 2015, a little drone flew 55km

across Appalachian Mountains to deliver medical supplies to a healthcare centre in Wise County, Virginia, USA. The place is not easily accessible and doctors usually stock up for a month at a stretch, leading to a lot of waste. This FAA-approved drone delivery has seeded hope in the minds of many of the area's citizens, who look forward to more such humanitarian drone missions.

From delivering medicines and commercial parcels to transporting organs, following clouds, spraying crops, shooting candid sports videos and surveying real estate, drones are attempting to become part and parcel of our lives, like cars and mobile devices once did. And soon these really might dot our skies, every day.

This prospect has put scientists and activists on full throttle. There is a lot of concern about the safety and privacy problems posed by these drones and the need for proper regulations to overcome these. Fortunately, there is a lot of research and development happening to make smaller, smarter, more useful drones. In this story, let us take a peek at some such won-drone-ful developments.

Drones turn into Mr Fix-Its

University of Leeds, UK, has undertaken to develop drones that can be used to autonomously fix city infrastructure such as mending potholes or changing streetlights. The idea is to have drones automatically survey the city's infrastructure, so problems can be spotted and fixed even before these are visible to the human eye. The project will take a three-pronged approach to solving this challenge.

One area of research, dubbed as Perch and Repair, will aim to develop drones that can perch atop high structures to fix stuff like streetlights.

Another dimension is Perceive and

Robo-Fly being developed by the US Army (Image courtesy: US Army)

